

TOSHIBA Field Effect Transistor Silicon N Channel Junction Type

2SK246

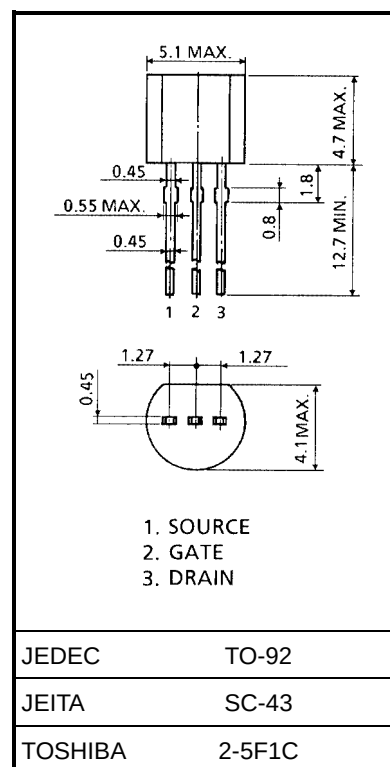
For Constant Current, Impedance
Converter and DC-AC High Input
Impedance Amplifier Circuit Applications

Unit: mm

- High breakdown voltage: $V_{GDS} = -50$ V
- High input impedance: $I_{GSS} = -1$ nA (max) ($V_{GS} = -30$ V)

Maximum Ratings ($T_a = 25^\circ\text{C}$)

Characteristics	Symbol	Rating	Unit
Gate-drain voltage	V_{GDS}	-50	V
Gate current	I_G	10	mA
Drain power dissipation	P_D	300	mW
Junction temperature	T_j	125	$^\circ\text{C}$
Storage temperature range	T_{stg}	-55~125	$^\circ\text{C}$



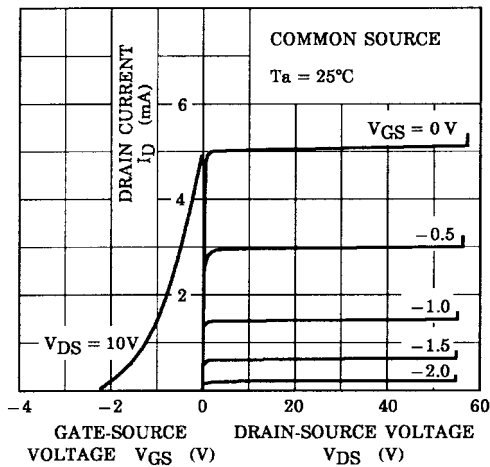
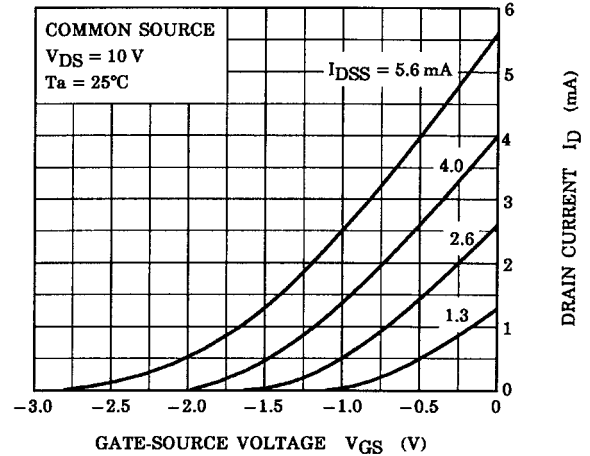
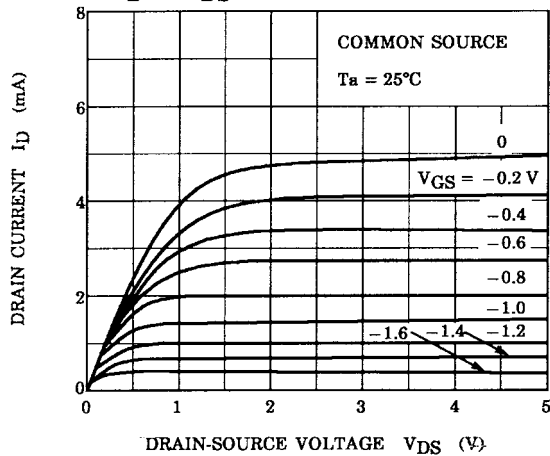
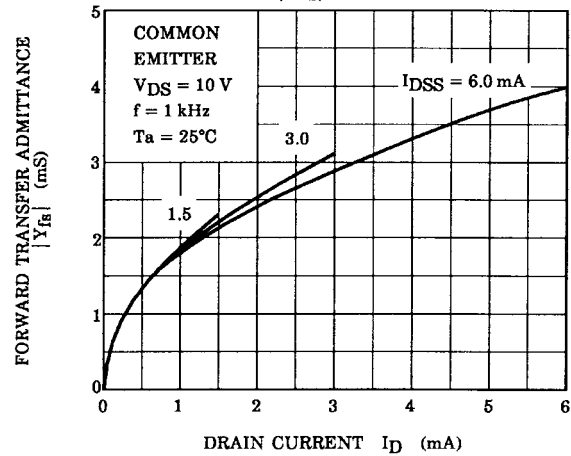
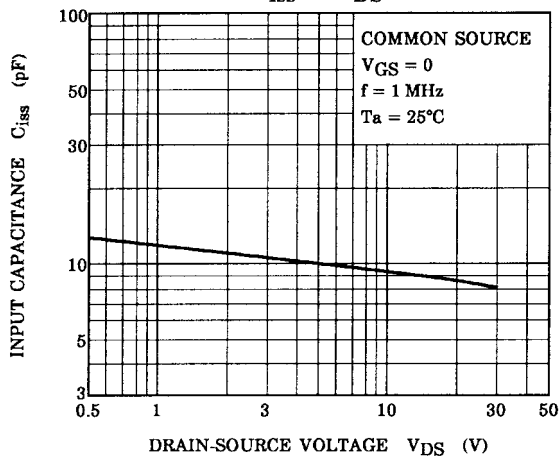
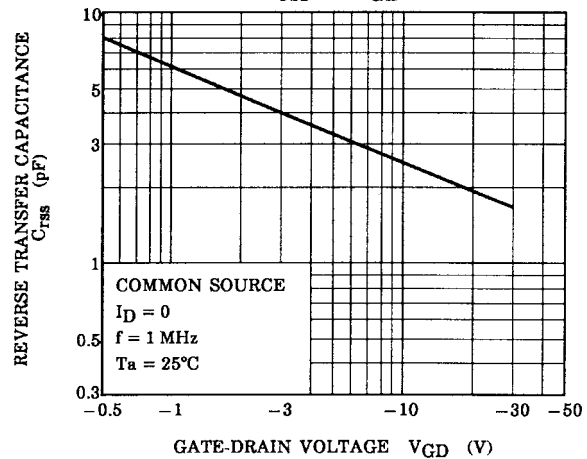
Weight: 0.21 g (typ.)

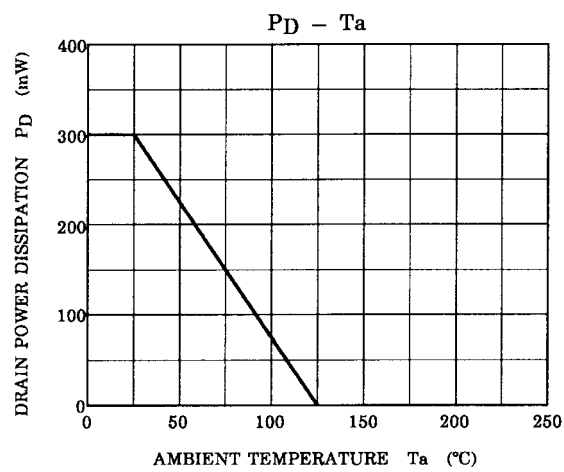
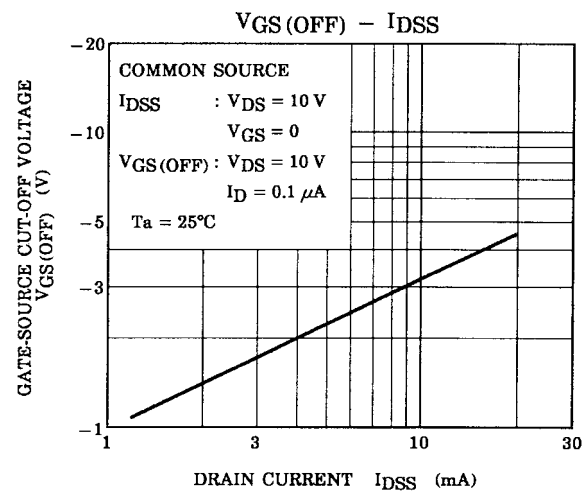
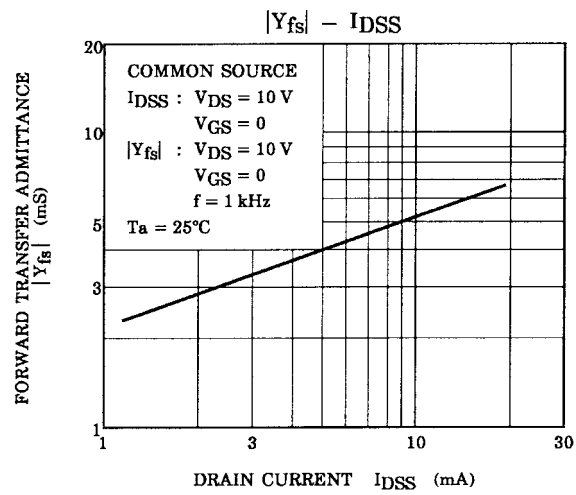
Electrical Characteristics ($T_a = 25^\circ\text{C}$)

Characteristics	Symbol	Test Condition	Min	Typ.	Max	Unit
Gate cut-off current	I_{GSS}	$V_{GS} = -30$ V, $V_{DS} = 0$	—	—	-1.0	nA
Gate-drain breakdown voltage	$V_{(BR) GDS}$	$V_{DS} = 0$, $I_G = -100$ μA	-50	—	—	V
Drain current	I_{DSS} (Note)	$V_{DS} = 10$ V, $V_{GS} = 0$	1.2	—	14	mA
Gate-source cut-off voltage	$V_{GS (OFF)}$	$V_{DS} = 10$ V, $I_D = 0.1$ μA	-0.7	—	-6.0	V
Forward transfer admittance	$ Y_{fs} $	$V_{DS} = 10$ V, $V_{GS} = 0$, $f = 1$ kHz	1.5	—	—	mS
Input capacitance	C_{iss}	$V_{DS} = 10$ V, $V_{GS} = 0$, $f = 1$ MHz	—	9.0	—	pF
Reverse transfer capacitance	C_{rss}	$V_{DG} = 10$ V, $I_D = 0$, $f = 1$ MHz	—	2.5	—	pF

Note: I_{DSS} classification Y: 1.2~3.0 mA, GR: 2.6~6.5 mA, BL: 6~14 mA

STATIC CHARACTERISTICS

 $I_D - V_{GS}$  $I_D - V_{DS}$ (LOW VOLTAGE REGION) $|Y_{fs}| - I_D$  $C_{iss} - V_{DS}$  $C_{rss} - V_{GD}$ 



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